STA302 Final Project

Introduction

With over 100 million tracks and 761 million users in 184 markets, Spotify is the most popular audio subscription streaming service globally (Spotify, 2026) among artists and listeners alike. TikTok and YouTube are both popular social media sites, the former being centered around short-form content ("TikToks"), and the latter a video-sharing platform where music artists upload music videos to promote the track. For any musical artist attempting to make a "hit," understanding the importance of cross-platform promotion is integral. In this study, we aim to answer the question: What is the relationship between the stream accumulation of a song on Spotify and its engagement metrics on Tiktok and YouTube, and does this relationship significantly differ across different release decades?

Literature Review

To ground this inquiry, we examine existing peer-reviewed literature. A study by Yee & Raheem (2022) used a machine learning algorithm to predict a song's popularity using YouTube engagement metrics and Spotify audio features (i.e., danceability, energy, key) across a dataset of tracks released in 2021. They found that the accuracy of their predictions significantly improved after including YouTube's engagement metrics in their model, highlighting the need for further investigation into cross-platform promotional effects across other dominant media like TikTok. Furthermore, Björklund et al. (2022) explored cross-platform streaming lifecycles, demonstrating that historical catalog tracks experience entirely distinct algorithmic recommendation paths and metrics over time compared to modern viral hits. This structural distinction provides the empirical basis for converting raw release dates into categorical predictors representing distinct release decades, allowing our model to isolate baseline streaming differences across musical eras.

Methodology

We will use multiple linear regression with interaction terms between the release decade and the qualitative predictors to analyze Spotify streams using TikTok views, likes, posts, and YouTube music video views and likes across different release decades. Multiple linear regression allows for the isolation of the relationship between individual social media engagement metrics and a song's Spotify streams while holding all other variables constant. This framework seamlessly accommodates both continuous numerical predictors and our categorical era predictor, prioritizing clear structural interpretability of music trends over blind prediction.

Data Description

The dataset we used in this project, "Most Streamed Spotify Songs 2024", was compiled by Nidula Elgiriyewithana (2024) and sourced from Kaggle. The original curator first collected the data by programmatically scraping public music web repositories and extracting metrics through official developer platform APIs (Spotify, YouTube, and TikTok) then formatted it into a CSV file available on Kaggle. While the original purpose of the dataset was simply to provide an all time ranking list of popular songs, our research proposal differs fundamentally by repurposing these metrics to build an explanatory framework through analyzing cross platform engagement structures.

The response variable is the number of Spotify Streams, a continuous quantitative metric measuring a track's total play count. Statistically, it features a massive right skew with a wide spread (Mean = 593.29M, Median = 419.01M, SD = 552.71M), driven by a small tier of hyper-viral tracks. It is highly suitable for multiple linear regression because it represents a continuous metric whose structural variation can be modeled mathematically as a linear combination of predictors.

Six predictors are selected to match our literature: the numbers of YouTube Views, YouTube Likes, TikTok Posts, TikTok Likes, TikTok Views, and the modified categorical variable Release Decade. The raw release dates were converted into a three-level categorical factor: 2020s, 2010s, and Pre-2010s. Grouping older dates this way preserves model degrees of freedom while also accounting for baseline shifts across music eras. All continuous predictors display heavy right skews and large standard deviations, reflecting the industry's "superstar effect". These distributions are highly relevant to our research question, as they allow us to see if engagement metrics scale linearly with streams or if baseline constants change depending on a track's release era. A summary of the predictors and response is given in the following table below.

Table 1

Summary Statistics of Predictor and Response Variables with n = 2778

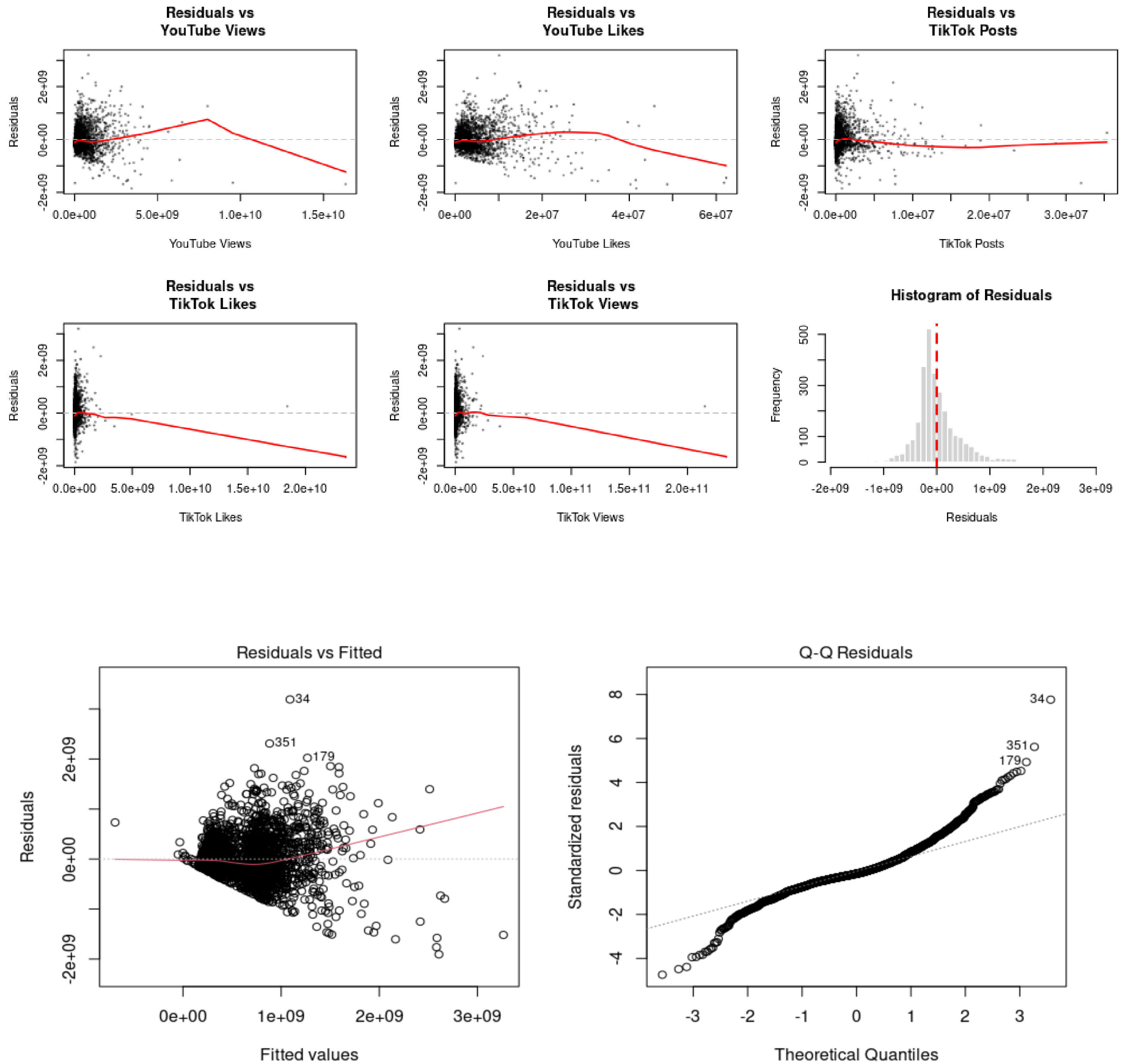
Variable	Mean	Std. Dev.	Median	Min	Pctl. 25	Pctl. 75	Max
Spotify Streams (Millions)	593.29	552.71	419.01	0.01	189.58	819.62	4281.47
YouTube Views (Millions)	499.71	755.93	239.67	0.28	97.89	622.78	16322.76
YouTube Likes (Millions)	3.8	5.1	2.14	0.01	0.86	4.72	62.31
TikTok Posts (Thousands)	819.31	2017.14	186.48	0	44.09	722.14	35323.8
TikTok Likes (Millions)	122.82	609.8	33.43	0	9.29	105.63	23474.22
TikTok Views (Millions)	1235.8	6507.4	316.04	0	89.44	1009.27	233232.31
Release Decade: Pre-2010s	2%						
Release Decade: 2010s	38.7%						
Release Decade: 2020s	59.4%						

Preliminary Results

A preliminary multiple linear regression model with interaction terms between the categorical and continuous predictors was fitted to estimate a song's total Spotify streams based on its cross-platform engagement metrics and release era. The reference group is Pre-2010s tracks with all continuous engagement metrics equal to zero. The intercept of approximately 1.18 billion streams represents the estimated baseline for such a track, sensibly large given decades of play accumulation. Among the numerical predictors, YouTube Likes shows the clearest positive relationship: each additional like is associated with approximately 85.08 more streams for a Pre-2010s track, though this weakens for 2010s (net ≈ 51.97) and 2020s tracks (net ≈ 13.35), suggesting YouTube likes are a stronger signal for older catalog music. YouTube views shift from negative for Pre-2010s (−0.432 per view) toward positive for 2020s tracks (+0.487), implying viewership-driven discovery matters more for newer songs. The TikTok predictors show near-zero main effects and negligible interaction terms across all decades, contributing little explanatory power once YouTube metrics and release era are accounted for. Holding all engagement metrics constant, 2010s tracks are estimated at approximately 545.1 million fewer streams than Pre-2010s, and 2020s tracks approximately 994.7 million fewer, aligning with Björklund et al.'s (2022) finding that catalog tracks follow distinct streaming lifecycles.

Figure 1

Residual Plots for Preliminary Model



Diagnostic plots (Figure 1) reveal violations of all regression assumptions. Linearity is violated, as the red lines curve across all Residuals vs. Predictor plots. Homoscedasticity is violated, with residuals fanning out as fitted values increase in their respective scatterplots. Normality is violated, as the Q-Q plot curves sharply at both tails and the histogram shows severe right skew. Independence of observations is also suspect, as viral trends likely induce clustering among observations. These violations motivate a possible Box-Cox transformation of Spotify Streams in the final model.

Model Selection

Transformations

To address the severe violations of homoscedasticity and normality identified in the preliminary diagnostics, a Box-Cox transformation was applied onto the response Spotify Streams. Maximum likelihood estimation obtained the optimal parameter of $\lambda \approx 0.3434$, rounded to $\lambda = \frac{1}{3}$ for theoretical parsimony and interpretive consistency, justifying a cube root transformation. Furthermore, a log transformation was applied to each predictor since the predictor data is heavily right-skewed, as reflected in Table 1. That is, for each continuous predictor, $\bar{x} \gg Med$, where $\bar{x}$ is the sample mean and Med is the sample median, thus dampening the effects of large predictor values on our model.

Leverage Points, Outliers, and Influential Points

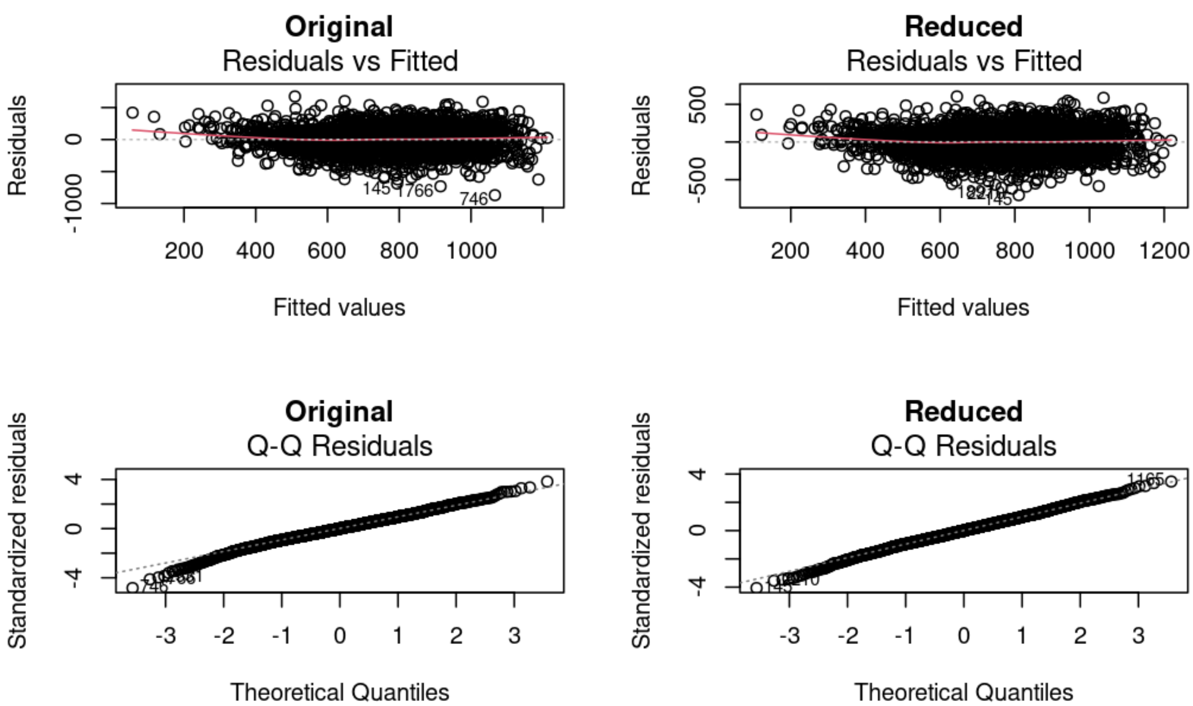
After transformation, leverage points were identified using hat values with a threshold of $2(\frac{p+1}{n})$, where p is the number of predictors and n is the sample size. A total of 134 leverage points were identified. With unusual predictor values, these points can exert disproportionate influence on the model, though leverage points will not necessarily distort the model and are still worth cross-referencing with outliers and influential points.

Standardized residuals r were used to identify outliers using the threshold of $|r| > 4$, which was chosen due to the large sample size. Two outliers were recognized. After inspection, we concluded that both outliers were legitimate observations and thus should be included.

Influential points were identified using Cook's Distance D_i, $DFFITS$, and $DFBETAS$, with thresholds of $\frac{4}{n}$, $2\sqrt{\frac{p+1}{n}}$, and $\frac{2}{\sqrt{n}}$ respectively. Such thresholds were chosen due to the large sample size. A total of 146 points were flagged that had high D_i, 147 points were flagged with high $|DFFITS|$, and 62 points were flagged with high $|DFBETAS|$. 44 points appeared in all three categories, and are therefore the most problematic. The transformed model was fit with and without these 44 points to determine their overall influence on the model.

Figure 2

Comparison of Residual Plots of the Original Transformed Model and the Reduced Transformed Model



In Figure 2 above, the Residual vs. Fitted plot for the reduced model suggests that the reduced model satisfies the constant variance assumption moderately more than the original. Furthermore, the new Q-Q plot, although subtle, displays a straighter line than the Q-Q plot for the original model with a lighter left tail and right tails, indicating the normality assumption for errors is satisfied better within the reduced model. The reduced model also had a ~0.016 increase in R^2_{adj} and a ~2% decrease in both AIC and BIC relative to the original model. Since these are marginal refinements, an inspection of each point is necessary to determine if the exclusion of any points is justifiable. Of the 44 most problematic influential points, a vast majority exhibited clear anomalies, prompting a removal of all 44 points. Some songs had implausibly high TikTok engagement metric values, possibly being data input errors, or indicating a song's virality mismatch across different platforms, and thus are statistically non-representative. Several tracks were live performances or heavily regional, with cross-platform metrics significantly differing from standard releases. Some Pre-2010s songs Spotify Streams also do not increase proportionally to TikTok engagement metrics when compared to more recent songs (which make up the vast majority of the data). A small minority of tracks (6) were genuinely high-performing songs across all platforms with no clear anomalies, and their exclusion in this model is recognized as a limitation.

Variable Selection

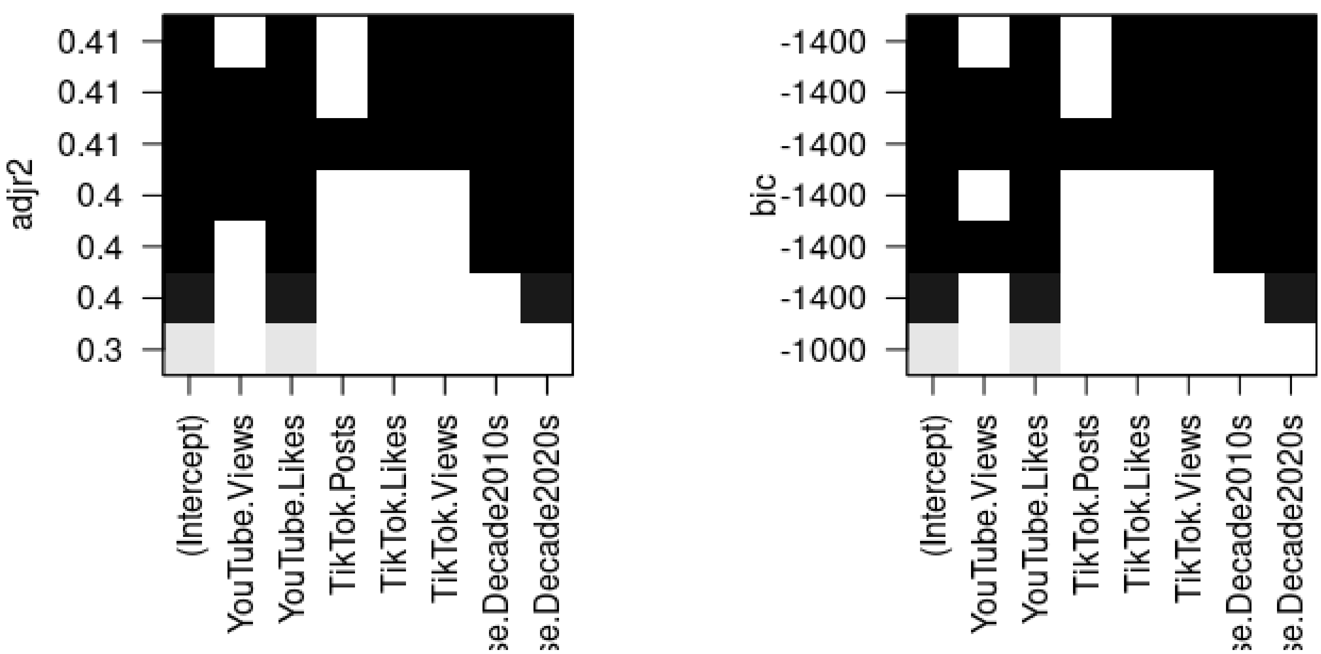
The preliminary model included interaction terms between the categorical predictor Release Decade and each continuous predictor. A partial F-test comparing the interaction model with the main-effects model indicated a statistically significant improvement in fit when interaction terms were included. However, further inspection of the individual interaction coefficients revealed high p-values, indicating a lack of statistical significance. Furthermore, the difference in adjusted R-squared was minimal, with the interaction model improving the value by only 0.0069. Given the negligible improvement in explanatory power and increased complexity, the interaction terms were removed from the final model.

When performing variable selection, the Bayesian Information Criterion was used to punish more complex models to increase the interpretability and transparency of the model since we had a high sample size. We used best subset selection because it was practical to use in this scenario; we have 6 potential predictors so best subset selection only needs to compare 2^6, or 64 models. Based on Figure 3 depicted below, we decided to exclude Youtube Views and TikTok Posts from our predictors, as both R^2_{adj} and BIC graphs showed that the best model excludes those predictors. Thus, we are left with Youtube Likes, TikTok Likes, TikTok Views, and our categorical predictor Release Date.

VIF analysis indicated severe multicollinearity between TikTok Likes and TikTok Views, with $GVIF^{\frac{1}{2Df}}$ values of over 5.5 for both. We considered two models: one containing TikTok Views and the other TikTok Likes. The model containing TikTok Likes had a higher R^2_{adj} (0.4875286 compared to 0.477025), along with lower AIC (36170.45 compared to 36225.92) and BIC (36205.93 compared to 36261.40) values. Thus, TikTok Views was removed from the final model and TikTok Likes was kept.

Figure 3

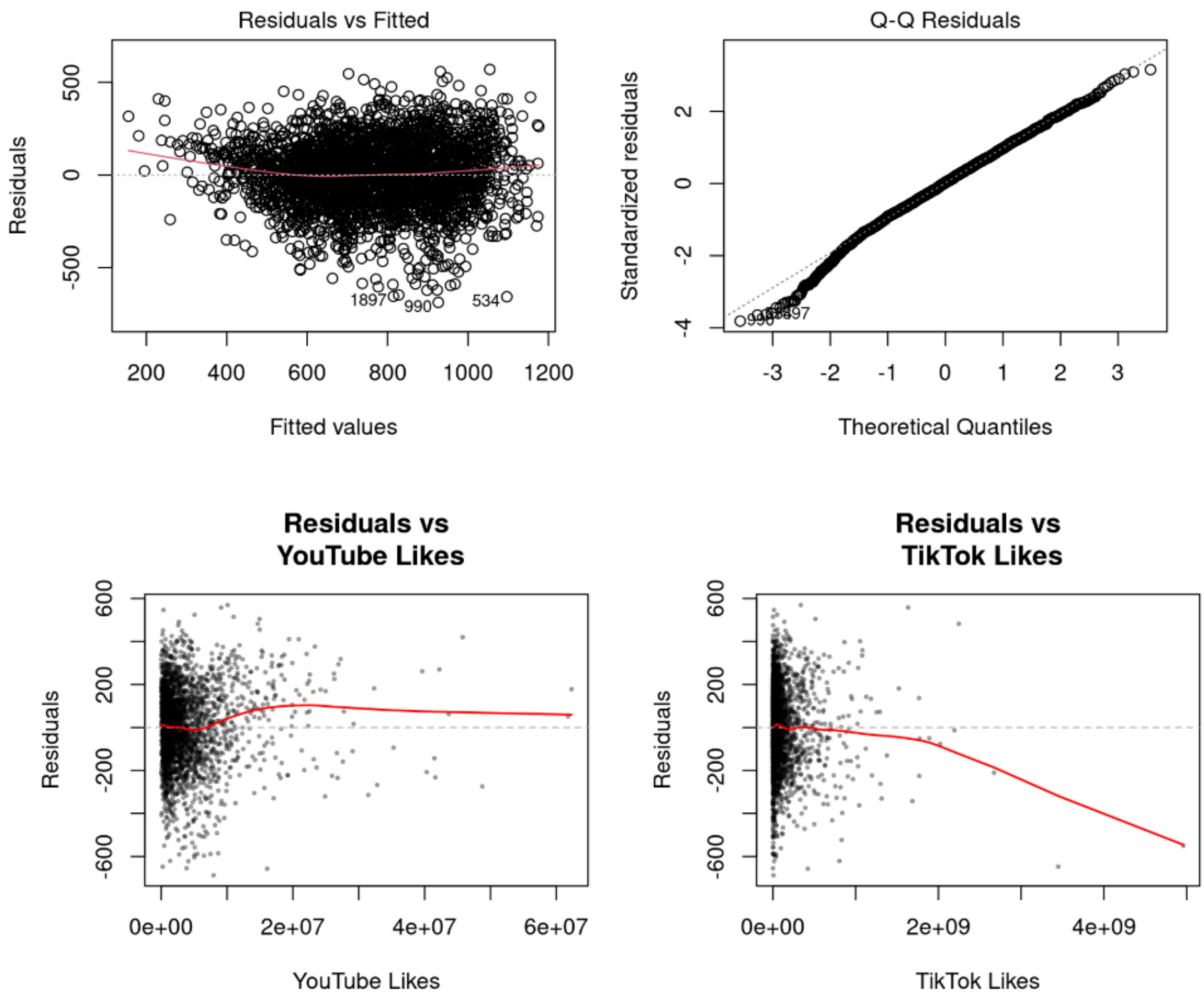
Best Subset Selection Results



Improvements upon Preliminary Model

Figure 4

Residual Plots after Transformation and Variable Selection



Overall, our new model significantly improves on our preliminary models by better satisfying the linear assumptions. This can be seen in Figure 4, since the Residual vs. Fitted plot does not display the distinct fan-shaped scatterplot as recognized in Figure 1. Furthermore, excluding the two predictors YouTube Views and TikTok posts prevents overspecifying and overfitting the model; including these predictors would result in a model with insufficient generalization ability. However, while the Residual vs. Predictor plots have improved, there are still glaring homoscedasticity and linearity issues. We recognize this as a limitation of our model.

The Final Model

The final refined model consists of the response variable, the cube root of Spotify Streams, and 3 predictors, YouTube Likes, TikTok Likes, and Release Decade, each continuous predictor log transformed.

Final Model Inference and Results

Table 2

Summary of Final Model with 95% Confidence Intervals

Variable	Estimate	Std. Error	CI Lower	CI Upper	t-statistic	p-value
(Intercept)	-645.054	62.132	-766.884	-523.224	-10.382	<2e-16
log(YouTube.Likes)	77.244	3.407	70.564	83.923	22.675	<2e-16
log(TikTok.Likes)	28.154	1.753	24.717	31.591	16.062	<2e-16
Release.Decade2010s	-99.211	34.541	-166.940	-31.482	-2.872	0.0041
Release.Decade2020s	-258.746	34.706	-326.799	-190.693	-7.455	1.20e-13

Interpretation of Coefficients

Since the final model incorporates a natural log transformation on the continuous predictors and also incorporates a cube root transformation ($\lambda = ⅓$) on the response variable, the regression slopes do not represent linear increments within each raw variable. Instead, they reflect linear relationships on a transformed scale, interpreted by holding all other variables constant. The model's intercept coefficient ($\hat{\beta}_0 = -645.054$, $p <$ 2e-16) represents the predicted cube root of the mean number of Spotify streams for a track in our reference category (Pre-2010s) when $log(YouTube.Likes)$ and $log(TikTok.Likes)$ are zero (that is, when $YouTube.Likes = 1$ and $TikTok.Likes = 1$). Since negative streaming volume is empirically impossible, there is no practical interpretation of this coefficient in relevance to the research question, although this suggests the model mathematically requires a very negative intercept to produce the extremely high predicted values of the observations seen in the data. The coefficient of $log(YouTube.Likes)$ demonstrates a robust positive association with streaming volume ($\hat{\beta}_1 = 77.244, p <$ 2e-16). For instance, if the number of a song's YouTube likes double, the expected cube root of the number of Spotify streams increases by $77.244 \times log(2) = 53.54$ units, suggesting that long-form video appreciation is a powerful indicator of sustained audio streaming success. β_2, the coefficient of $log(TikTok.Likes)$, demonstrates that short-form engagement also exhibits a highly significant positive relationship with play counts ($\hat{\beta}_2 = 28.154, p <$ 2e-16). Doubling the number of a track's TikTok likes corresponds to an estimated $28.154 \times log(2) = 19.51$ unit increase in the response variable, reinforcing TikTok as a distinct and functional pipeline for driving the increase in global music consumption. When comparing the coefficients of $log(TikTok.Likes)$ and $log(YouTube.Likes)$, note the coefficient of $log(YouTube.Likes)$ is significantly larger, suggesting the engagement in a specific artist's music videos is a stronger driver of expected Spotify streaming volume compared to likes on TikTok. The estimated coefficient of Release.Decade2010s ($\hat{\beta}_3$) implies that, while holding cross-platform metrics constant, 2010s tracks are associated with an expected decrease of 99.211 units ($p = 0.00411$) of the mean value of the cubed root of streaming volume relative to Pre-2010s catalog music. This negative intercept difference is even more pronounced for tracks released within this decade ($\hat{\beta}_4 =- 258.746, p = 1.2$e-13). That is, when controlling for all social media metrics, a 2020s release is predicted to drop by an even larger 258.746 units on the cubic root scale compared to the reference group. Both coefficients of the dummy variables lead to the implication that there is a positive correlation between a track's social media engagement metrics and its accumulated Spotify streaming volume, and there is likely an even greater dependence of 2020s songs on cross-platform promotion in comparison to 2010s songs.

Implications for the Research Question

The regression coefficients reveal a fundamental dynamic regarding our central inquiry: multi-platform viral metrics are critical, but the degree of their influence is inextricably linked to the track's release era. The increasingly negative parameters associated with modern decades suggest a substantial "accumulation effect." Specifically, older catalog tracks possess a structural head start; by existing in the market for several decades, they have established an exceptionally high baseline of cumulative stream volume before the era of social media. This is also suggested by the current circumstance that a majority of contemporary releases with Spotify streams rivaling those of legacy baseline songs achieve significantly higher engagement thresholds on TikTok and YouTube.

Additionally, there also lies a significance in the platform used to promote a track. Our interpretation of the regression coefficients reveal that YouTube music video appreciation is a stronger driver for Spotify streams than TikTok content appreciation. Since the interaction terms from the preliminary model were deemed insignificant and thus excluded in the final model, this hierarchy of media engagement is unlikely to differ across songs' decade of release. Thus, while overall multi-platform metrics influence the streaming numbers of legacy baseline songs less compared to contemporary songs, YouTube likes is the primary agent (over TikTok likes) in propelling streaming numbers of all songs, no matter the decade.

Comparison with Literature Findings

Our final analysis exhibits strong empirical alignment with the theoretical frameworks established in previous academic studies. In reference to social engagement frameworks, our results corroborate the primary conclusions of Yee & Raheem (2022), in that YouTube engagement metrics are a critical predictor of a song's popularity. The validation that log-transformed likes across both long-form video (YouTube) and short-form media (TikTok) are robust, independent predictors of total stream volume ($p <$ 2e-16) highlights the necessity of viewing streaming metrics through a multi-platform social lens. As for catalog lifecycle trends, the significant negative baseline shifts for the 2010s and 2020s eras mathematically support the music lifecycle theories proposed by Björklund et al. (2022). Modern hits follow a distinct trajectory, relying on immediate spikes in short-form engagement to drive discovery, whereas legacy tracks leverage decades of passive accumulation to maintain high stream totals despite having lower current social media interaction.

Model Performance Metrics

The fit of the final main-effects model demonstrates moderate explanatory performance and improved parsimony compared to the initial model architectures. The final model achieves an $R^2_{adj} = 0.4875$ ($R^2 = 0.4883$). This indicates that 48.75% of the variance in the cube root of Spotify streams is accounted for by the combination of our log-transformed engagement metrics and the release decade indicator, a moderate amount. Given there are an insurmountable number of factors that determine a song's popularity, this is to be expected. Still, this represents a meaningful improvement over the untransformed preliminary model, which had an $R^2_{adj} = 0.4455$. The final model is also significantly closer to being correctly specified, excluding extraneous terms, reducing multicolinearity by reducing 6 parameters to the 4 most meaningful, and the removal of problematic influential points. This is corroborated by the slight improvements in the AIC and BIC from the preliminary model to the final model, thus signaling the final model fits the data better even after accounting for complexity, and the removal of the TikTok Views predictor due to high multicollinearity with TikTok Likes. Furthermore, the Residual Standard Error (RSE) for this scale is 180.3 on 2729 degrees of freedom, an extremely significant improvement when compared to the massive RSE of the untransformed baseline (411,600,000). This reflects both an upgrade in model fit and a successful stabilization of the error distribution through the cube root and log transformations, thus satisfying homoscedasticity assumptions more effectively. The overall fit is also supported by a global F-statistic of 651 on 4 and 2729 degrees of freedom, yielding a model p-value of $p <$ 2.20e-16. Therefore, our findings are statistically significant, and the predictors, collectively, are meaningfully associated with the response. Overall, in combination with the model's R^2_{adj} value, our model has moderate, yet meaningful, explanatory power, and the F test confirms that this explanatory relationship is not due to chance.

Discussion and Conclusion

The central aim of this inquiry involved assessing the structural association between Spotify streaming accumulation and promotional engagement across the long-form (YouTube) and short-form (TikTok) digital ecosystems. Utilizing an optimized additive regression framework, our findings indicate that multi-platform user engagement metrics serve as highly robust and statistically significant predictors of global streaming volume.

Furthermore, our analysis identifies the release era of a track as a critical baseline moderator. This highlights a fundamental market divergence: modern frontline releases necessitate extreme levels of active digital interaction to remain competitive, whereas legacy catalog tracks benefit from a substantial baseline advantage established through decades of continuous market exposure.

Key Findings and Supporting Evidence

Our results provide three definitive insights that directly address the core research question regarding cross-platform synergies in the modern digital music ecosystem. By analyzing the interconnected nature of long-form video, short-form social engagement, and release eras, our predictive modeling offers a comprehensive understanding of how different promotional vectors influence overall performance on Spotify.

First is the domination of Youtube video engagement. Cumulative YouTube appreciation surfaced as the most influential continuous predictor ($\hat{\beta}_1 = 77.244, p <$ 2e-16). Controlling for all other variables, a doubling of YouTube likes is associated with an increase of 53.54 units in the expected cube root of Spotify streams. This robust relationship suggests that deep visual engagement on video-sharing platforms directly mirrors or propels active consumer consumption on audio streaming services. Unlike passive scrolling, seeking out and engaging with a full-length music video signifies a higher level of user intent and fan dedication, which naturally translates into sustained, repeated listening behavior on Spotify.

Second is the distinct role of short-form content virality, as TikTok likes also demonstrated a highly significant positive relationship with stream volume ($\hat{\beta}_2 = 28.154, p <$ 2e-16). Doubling TikTok likes corresponds to an estimated 19.51 unit increase in the cube root of Spotify plays. This implies that while TikTok functions as a vital, high-velocity discovery mechanism, its metric maps to a notably lower scalar shift than YouTube. We interpret this to mean that short-form virality acts as an effective directional pipeline for user attention, but it does not guarantee long-term audio retention.

Finally, we have to take into account the effect of historical accumulation. Categorical era coefficients revealed a steep negative shift for contemporary tracks, with baseline parameters dropping by 99.211 units for the 2010s and 258.746 units for the 2020s. This confirms that legacy music, or “classics”, holds a profound structural advantage in the modern streaming landscape. Older tracks have leveraged decades of cultural permeation to establish a highly resilient streaming floor. In stark contrast, modern frontline releases are uniquely dependent on acute social media spikes to offset their lack of historical presence. Because passive listening habits and automated playlists frequently default to recognizable catalog classics, new releases are forced to aggressively rely on cross-platform virality to compete for initial market share.

Practical Implications and Recommendations

The empirical findings detailed above extend far beyond theoretical observations, offering essential, data-driven practical implications for record labels, independent producers, artist managers, and digital marketers navigating an increasingly fragmented media landscape. By translating these insights into actionable frameworks, industry stakeholders can optimize resource allocation and maximize their return on investment across multiple digital touchpoints.

First is the concept of ecosystem-wide promotional stacking. Because our analysis identifies both YouTube and TikTok engagement as independent, statistically significant drivers of audio consumption, modern campaigns should adopt an interconnected "stacking" approach. Marketers should actively leverage viral, short-form triggers on platforms like TikTok to capture initial algorithmic discovery and capitalize on fleeting consumer attention. Simultaneously, they must invest in high-quality, long-form visual assets on YouTube to cultivate deeper audience loyalty, contextualize the artist's brand, and optimize the final conversion into sustained streaming behavior.

Next are the differentiated strategies for catalog compared to frontline music. Rights holders and managers of legacy catalog music benefit from a structural safety net; they can rely on predictable baseline streaming totals driven by nostalgia, established algorithmic trust, and passive placement in mood-based playlists. Conversely, frontline 2020s releases face a steep baseline deficit. Without the luxury of historical accumulation, new artists and modern releases require aggressive, sustained, and relentless digital engagement to break through the noise. They must actively manufacture cultural moments to overcome their inherent historical disadvantage, demanding a much higher velocity of ongoing, cross-platform content creation to secure a permanent foothold in today's attention economy.

Limitations of the Analysis

Despite the general robustness and predictive power of the global model, several structural limitations must be acknowledged to properly contextualize our findings.

The first constraint involves the exclusion of extreme outliers during diagnostic testing. Our rigorous data cleaning phase excluded 44 highly influential observations that violated standard statistical thresholds. While most of these were genuine anomalies, a distinct minority represented actual ultra-viral mega-hits. Excluding these specific data points inherently prevents the current model from accurately capturing the absolute upper bounds of massive pop culture phenomena.

Next is the constraint on platform exclusivity, as we deliberately limited our independent variables to YouTube and TikTok metrics. However, significant consumer interaction and music discovery consistently occur across other digital ecosystems, such as Instagram Reels and curated Apple Music playlists. Because these vital engagement vectors remain entirely unrepresented within this specific analytical framework, the model may understate a track's total digital reach.

Finally, the reliance on cross-sectional data heavily restricts our temporal insights. The collected dataset provides only a single, static snapshot in time, effectively preventing the model from tracking complex longitudinal dynamics. Consequently, we cannot measure critical time-based factors, such as viral decay rates or the specific speed at which a sudden social media spike transitions into stable, long-term audio streams.

Future Studies and Improvements

To address the aforementioned limitations, future research should implement the following technical refinements.

First, rather than discarding ultra-viral tracks to satisfy the restrictive constraints of traditional linear regression, future studies should actively embrace outlier data. Employing advanced modeling techniques like piecewise regression or generalized additive models (GAMs) would allow researchers to accurately capture the sudden, exponential behavioral shifts associated with extreme global virality. Retaining these blockbuster mega-hits is absolutely essential for properly modeling the highly skewed "winner-takes-all" economy that defines the modern music industry.

Furthermore, to effectively capture a track's dynamic lifecycle, future analyses must utilize continuous weekly or monthly updates rather than relying on static, cumulative cross-sectional files. By incorporating panel data and mixed-effects models, researchers could meticulously map the specific lag times between initial social media interactions and subsequent audio plays. This continuous temporal tracking is crucial for revealing true directional causality, measuring the precise decay rates of fleeting viral trends, and understanding how quickly a digital spike transitions into stable consumption.

Finally, future analytical frameworks should significantly broaden their scope by incorporating a wider spectrum of independent variables to capture the complete digital ecosystem. This expansion must include critical external drivers like algorithmic and editorial playlist placements, alongside intrinsic qualitative acoustic features such as track tempo, danceability, and energy. By synthesizing these fundamental sonic characteristics with cross-platform engagement metrics, predictive models can better account for unexplained residual variance and significantly minimize omitted variable bias.

Contributions

Rita Huang

- Set up the framework for the project (creating the documents, communication, github repository)
- Helped choose relevant predictors/response variable(s) for the project
- Helped format and write R code and helped write the preliminary results section
- Proofread entire proposal and fixed errors
- Performed box-cox and helped justify model selection
- Helped write the discussion and conclusion, reviewed final document

Jennifer Du

- Provided a peer-reviewed academic source
- Helped choose relevant predictors/response variable(s) for the project
- Wrote R code to clean the data, produce some residual plots, and perform analysis on different models
- Helped justify model selection with ANOVA and VIF analysis

Sophia Ning

- Provided a peer-reviewed academic source
- Helped choose relevant predictors/response variable(s) for the project
- Partly wrote and edited introduction
- Created statistical summary table
- Produced some residual plots
- Wrote problematic points section in model selection
- Partly wrote and edited final model inference section

Crystal Lam

- Wrote introduction
- Formatted APA bibliography
- Wrote Data Description
- Helped write preliminary results
- Proofread rmd file, edited and reordered code to fix compilation errors and formatting.
- Wrote draft for final model inference and results section
- Helped write draft for discussion and conclusion

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